CHAPTER 3.

FORMAL LOGIC IN PHILOSOPHY

BAHRAM ASSADIAN

This chapter discusses some philosophical issues concerning the nature of formal logic. Particular attention will be given to the concept of logical form, the goal of formal logic in capturing logical form, and the explanation of validity in terms of logical form. We shall see how this understanding of the notion of validity allows us to identify what we call formal fallacies, which are mistakes in an argument due to its logical form. We shall also discuss some philosophical problems about the nature of logical forms. For the sake of simplicity, our focus will be on propositional logic. But many of the results to be discussed do not depend on this choice, and are applicable to more advanced logical systems.

LOGIC, VALIDITY, AND LOGICAL FORMS

Different sciences have different subject matters: physics tries to discover the properties of matter, history aims to discover what happened in the past, biology studies the development and evolution of living organisms, mathematics is, or at least seems to be, about numbers, sets, geometrical spaces, and the like. But what is it that logic investigates? What, indeed, is logic?

This is an essentially philosophical question, but its answer requires reflection on the status and behavior of logical rules and inferences. Textbooks typically present logic as the science of the relation of *consequence* that holds between the premises and the conclusion of a **valid argument**, where an argument is valid if it is not possible for its premises to be true and the conclusion false. If logic is the science of the relation of consequence that holds between the premises and the conclusion of a valid argument, we can say that logicians will be concerned with whether a conclusion of an argument is or is not a consequence of its premises.

Let us examine the notion of validity with more care. For example, consider the following argument:

1. If Alex is a sea bream, then Alex is not a rose.
2. Alex is a rose.

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3. / ∴ Alex is not a sea bream.

It can be shown that it is not possible for (1) and (2) to be true yet (3) false. Hence, the whole argument is valid. For convenience, let us represent each sentence of the argument into the standard propositional logic, which aims to analyze the structure and meaning of various propositions. To do this, we must first introduce the language of our logic.

The alphabet of propositional logic contains letters standing for sentences: *A*, *B*, *C*, and so on. For example, we can translate "Alex is a rose" by just using *B*. Similarly, we can use *S* to translate "I would love to smell it." The alphabet of propositional logic contains other symbols known as **logical connectives**. One is a symbol for "not" or *negation* $(\neg)$. When we say that Alex is *not* a rose, we, in effect, say that it is not the case that Alex is a rose. If we translate "Alex is a rose" by *B*, we translate "Alex is not a rose" as "$\neg B$." Another is a symbol $(\rightarrow)$ for conditional sentences of the form "if … then …." For example, we can translate "If Alex is a rose, then I would love to smell it" as "$B \rightarrow A$." When we say that if Alex is a rose, then I would love to smell it, we say something conditional: on the condition that Alex is a rose, I would love to smell it. In general, a conditional sentence has two components. We call the first component the *antecedent*, the second component the *consequent*, and the whole proposition a **conditional**. The language of our logic also includes "and" $(\wedge)$, otherwise known as *conjunction*, and "or" $(\vee)$, otherwise known as *disjunction*. But in this chapter, we shall only deal with negation and conditional.

Thus, if we use *A* for "Alex is a sea bream," we can represent (1) with $A \rightarrow \neg B$, and represent our above argument (1)-(3) as follows:

1. $A \rightarrow \neg B$
2. B

---

3. $/ \therefore \neg A$

But, recall, our aim was to examine why this argument, if at all, is valid. The mere representation of "not" by "$\neg$ " and "if … then" by "$\rightarrow$" will not be sufficient to verify the validity or invalidity of a given argument: we also need to know what these symbols and the propositions they express *mean*. But how can we specify the *meaning* of "neg " and "$\rightarrow$"?

It is plausible to say that if *A* is true, then its negation is false, and vice versa. For example, if "Alex is a rose" is true, then "Alex is not a rose" is false. This gives us the meaning of "$\neg$". We can represent this information about the meaning of negation in terms of a *truth-table* in the following way (with T symbolising *true*, and F *false*):

Truth table for negation

A	$\neg A$
T	F
F	T

Here, we can read each row of the truth-table as a way the world could be. That is, in situations or possible worlds where *A* is true (for example, where Alex is indeed a sea bream), $\neg A$ is false (it is false that Alex is a sea bream);

and vice versa. Thus construed, a truth-table gives us the situations in which a proposition such as A is true, and those in which it is false. In addition, it tells us in what situations $\neg A$ is true, and in what situations it is false.

In a similar way, we can specify the meaning of "$\rightarrow$" by specifying the situations in which conditional propositions of the form "$A \rightarrow B$" are true or false. Here is the standard truth-table for "$\rightarrow$":

Truth table for material conditional

A	B	$A \rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

As can be seen, there is only one row in which $A \rightarrow B$ is false; i.e. the second row in which the consequent is false, but the antecedent is true. As the first row tells us, if both A and B are true, then so is $A \rightarrow B$. Further, the third and fourth rows tell us that if the antecedent is false, then the whole conditional is true, regardless of whether the consequent is true or false. Hence, all conditionals with false antecedents are true.

But how is it possible for a conditional to be true if its antecedent is false? Here is one suggestion to answer this question: if your assumption is false, then you can legitimately conclude whatever you would like to. For example, if you assume that Amsterdam is the capital of England, you can legitimately conclude anything whatsoever; it does not matter whether it's true or false. Thus, from the assumption that Amsterdam is the capital of England, you can conclude that Paris is the capital of France. You can also conclude that Paris is the capital of Brazil.

We can see that one important piece of information that truth-tables convey concerns how the truth or falsity of complex sentences such as $A \rightarrow B$ and $\neg A$ depends on the truth or falsity of the propositional letters they contain: the truth or falsity of $A \rightarrow B$ depends solely on the truth or falsity of A and of B. Similarly, the truth or falsity of $\neg A$ depends solely on that of A.

Now we are in a position to verify whether our argument (1)-(3) is valid or not. And, as we shall see in a moment, the validity or invalidity of an argument depends on the meaning of the logical connectives (such as "$\rightarrow$" and "$\neg$") which is specified by the corresponding truth-tables. In other words, if the truth-tables of these connectives were different to what they actually are, we would have a different collection of valid arguments.

We defined an argument as valid if it is not possible for its premises to be true and the conclusion false. By designing a truth-table, we can see under what conditions the premises $(A \rightarrow \neg B, B)$ and the conclusion $(\neg A)$ of our argument (1)-(3) are true or false:

Truth table for argument (1)-(3)

A	A	$A \to \neg B$	B	$\neg A$
T	T	F	T	F
T	F	F	F	T
F	T	T	T	T
F	F	T	F	T

Since in the above truth-table, there is no row in which the premises $(A \to \neg B, B)$ are true and the conclusion $(\neg A)$ false, the argument is valid. The only row in which the premises are both true is the third row, and in that row the conclusion is also true. In other words, there is no world or situation in which (1) and (2) are true, but (3) is not. This just means that the argument is valid.

Now, consider the following argument:

4. If Alex is a tiger, then Alex is an animal.
5. Alex is not a tiger.

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6. $/ \therefore$ Alex is not an animal.

There are situations in which the argument works perfectly well. For example, suppose that Alex is not a tiger but is, in fact, a table. In this case, Alex would not be an animal, either. And thus, the sentences (4), (5), and (6) would be true. But this is not *always* the case, for we can imagine a situation in which the premises are true but the conclusion false, such as where Alex is not a tiger but is, in fact, a dog. Thus, by imagining the situation just described, we would have produced a counterexample: in this situation, (6) would be false, and hence it would not be a consequence of (4) and (5). The argument is invalid.

That the argument is invalid can also be verified by the method of truth-tables. For we can find a situation in which (4) and (5) are both true and yet (6) false. That is, in the truth-table, if we represent (4) as $C \to D$, (5) as $\neg C$, and (6) as $\neg D$, there will be at least one row in which the premises are true and the conclusion false (which row is that?):

Truth table for argument (4)-(6)

C	D	$C \to D$	$\neg C$	$\neg D$
T	T	T	F	F
T	F	F	F	T
F	T	T	T	F
F	F	T	T	T

We said that logicians are concerned with validity or invalidity of arguments, and we proposed the method of truth-tables for undertaking this task. But which arguments are valid, and which are not? It is here that the notion of **logical form** emerges. Suppose that a logician embarks on the ridiculous task of recording each and every valid argument. In this case, she would surely record that (1)-(3) is valid. Now, suppose she faces the following argument:

7. If Alice is reading Hegel, she is not frustrated.
8. Alice is frustrated.

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9. $/ \therefore$ Alice is not reading Hegel.

To see whether this argument is valid or not, she can rewrite each sentence of the argument in her logical language: Alice is reading Hegel (P); Alice is frustrated (Q); and, if Alice is reading Hegel, then Alice is not frustrated) $(P \to \neg Q)$. She can then design a suitable truth-table, and check whether there is any row or situation in which the premises are both true and the conclusion false. Since there is no such row (why?), she will correctly announce that the argument is valid.

But it is obvious that in order to check the validity of (7)-(9), our logician did not need to go to this effort. It would suffice if she just noted that the two arguments (1)-(3) and (7)-(9), and their respective truth-tables, are to a great extent *similar*; they have the same *form*. In fact, their only difference is that in the first, the letters *A* and *B* have been used, and in the second they have been substituted for *P* and *Q*, respectively. The logical connectives $\longrightarrow$ and $\neg$ have not changed.

To see the point, let us translate each argument into the language of propositional logic we introduced above:

1. $A \to \neg B$
2. B

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3. $/ \therefore \neg A$

7. $P \to \neg Q$
8. Q

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9. $/ \therefore \neg P$

The two arguments have something in *common*. Let us say that what they have in common is their *logical form*. As you can see, the logical connectives of the arguments have not changed. Since the two arguments have the same form, if one is valid, then the other must be valid, too. More generally, all arguments of this same form are valid. The liberating news is that our logician does not need to embark on the exasperating task of checking the validity of each and every argument separately. For if she already knows that a given argument is valid, and if she can also show that another argument has the same form as the first one, then she can be sure that the second argument is valid without having to design its truth-table.

We said that an argument is valid if it is not possible for the premises to be true and the conclusion false. Now, we can say that every argument which shares its form with a valid argument is also valid, and consequently, every argument which shares its form with an invalid argument is also invalid.[1] It is in this sense that the idea of logical form can be used to establish the (in)validity of arguments. For example, suppose that we want to check the validity of the following argument:

10. If Alice is reading Russell, then Alice is thinking of logic.
11. Alice is not reading Russell.

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12. / ∴ Alice is not thinking of logic.

As soon as we see that (10)-(12) has the same form as (4)-(6), which we already know to be invalid, we can be assured that the former is also invalid without having to construct its truth-table.

Thus, we can see that understanding the notion of validity in terms of logical form allows us to identify various *formal fallacies*. For example, the argument (10)-(12) is an instance of the fallacy of *denying the antecedent*. Thus, every argument which shares its form with (10)-(12) is also invalid.

There are three further questions we may ask about logical forms: (i) How can we "extract" the logical form from arguments which they share? That is, how can we show that various arguments are instances of a common logical form? (ii) What is the nature of a logical form? Is a logical form a *thing*, and if so, what sort of thing is it? (iii) Does each argument have only one logical form? In the following three sections, we shall talk about these three questions, respectively.

1. It is more accurate to say that every argument which shares its form with an invalid argument is also invalid *within that logic,* but not necessarily for every logic. For example, in propositional logic,

    1. All men are mortal
    2. Socrates is a man
    3. / ∴ Socrates is mortal

    is of the same logical form as:

    4. All men are immortal
    5. Socrates is a man
    6. / ∴ Socrates is mortal

    Both of these arguments can be translated as follows:

    i. P
    ii. Q
    iii. / ∴ R

    But (4)-(6), as opposed to (1)-(3), is invalid, for if all men are immortal and Socrates is a man, then Socrates is immortal. Thus, in propositional logic, both of these arguments have the same logical form, even though, from the perspective of a more expressive logic, such as first-order logic, which explains the role that quantifiers such as "all" and "some" play within arguments, only the first is valid. Thus, every argument which shares its form with a valid argument is valid within that logic, but not necessarily across the board.

EXTRACTING LOGICAL FORMS

Let us, again, consider the arguments (1)-(3) and (7)-(9) which seem to share one and the same logical form. How can we *show* that they have a common logical form? First, we should represent them in logical symbols:

1. $A \rightarrow \neg B$
2. B

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3. $/ \therefore \neg A$

7. $P \rightarrow \neg Q$
8. Q

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9. $/ \therefore \neg P$

To see what these two arguments have in common, we must abstract away from (or ignore or leave aside) the specific contents of their particular premises and conclusions, and thereby reveal a general form that is common to these arguments. For example, we must ignore whether Alex is or is not a rose; all that matters is to replace "Alex is a rose" with *B*. In this sense, to obtain or extract the logical form of an argument, we must abstract from the content of the premises and the conclusion by regarding them as mere place-holders in the form that the argument exhibits. As you may have noted, we do *not* extract away the content of the *logical connectives*. It is an important question as to why we do not abstract away from the logical connectives. The basic thought is that their meaning constitutes an important part of the logical form of an argument, and thereby in determining its (in)validity.

To talk about logical forms, we shall use the lowercase Greek letters such as $\alpha, \beta, \gamma,$ and δ. For example, we can represent the logical form that (1)-(3) and (7)-(9) share as follows:

i. $\alpha \rightarrow \neg\beta$
ii. β

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iii. $/ \therefore \neg\alpha$

An analogy may help here: In mathematics, we think about *particular* arithmetical propositions such as "$1 + 2 = 2 + 1$" and "$0 + 2 = 2 + 0$." But when we want to *generalize*, we use formulas that contain *variables*, and not *specific* numbers. For example, "$x + y = y + x$" expresses something general about the behaviour of the natural numbers. Whatever natural numbers *x* and *y* stand for, "$x + y = y + x$" remains true. The same goes with the variables $\alpha, \beta, \gamma,$ and δ, which enable us to talk in a general way about the premises and conclusion of arguments. Whatever meaning α and β are given, that is, whatever propositions they express, (i)-(iii) remains valid, and so do all of its instances, such as (1)-(3) and (7)-(9).

As mentioned above, extracting a certain logical form allows us to talk, in a general way, about premises and conclusions of arguments. It does not matter what specific objects and properties—what specific subject matter—they talk about. And this leads us, again, to our initial concern about the real subject matter of logic:

> Form can thus be studied independently of subject-matter, and it is mainly in virtue of their form, as it turns out, rather than their subject-matter that arguments are valid or invalid. Hence it is the forms of argument, rather than actual arguments themselves, that logic investigates. (Lemmon 1971, 4)

According to this conception of logic, logicians are in a position to evaluate the validity of an argument, even if they do not strictly understand the content of the claims within the argument, nor under what conditions they would be true. Whether or not the claims within arguments are true, therefore, is not a matter for logic. Instead, what logic does is to explore the *logical forms* of arguments, and thereby establish their (in)validity.

THE NATURE OF LOGICAL FORMS

In this and the next section, we will look into more philosophical matters. In this section, we shall discuss our second question: what is the nature of a logical form? The question about the nature of logical form is reminiscent of the ancient question about the nature of universals. All red roses have something in common; they all share or instantiate something. But what is that thing, if it is a thing at all? Is the property of *being red* akin to a Platonic universal that exists independently of the red roses that instantiate it? Or is it like an Aristotelian universal whose existence depends on the existence of the particular roses? Perhaps, it does not have any existence at all; it is nothing more than a name or a label that we use to talk about red roses. We can ask exactly the parallel questions about logical forms: What is it that all valid arguments of the same form share or instantiate? Is it an entity in the world, or a symbol in language, or a mental construction formed and created by us?

Assuming that logical forms exist, what are they? There are, generally speaking, two lines of thought here. According to the first, logical forms are *schemata*, and hence, are linguistic entities. According to the second, logical forms are *properties*: they are extra-linguistic entities, akin to universals. They are what schemata express or represent. (An analogy may help here: The expression "is happy" is a predicate; it is a linguistic item. But it expresses an extra-linguistic entity, such as the property of *being happy*.)

Identifying logical forms with schemata appears to be quite intuitive. But it leads to a fallacy. As Timothy Smiley points out, the fallacy lies in "treating the medium as the message" (Smiley 1982, 3). Consider the logical form of (1)-(3):

i. $\alpha \rightarrow \neg\beta$
ii. β

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iii. $/ \therefore \neg\alpha$

You may like, with equal right, to identify the logical form of (1)-(3) with:

iv. $\gamma \rightarrow \neg\eta$
v. η

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vi. $/ \therefore \neg\gamma$

And yet another logician may prefer to capture its logical form with a distinct set of variables:

vii. $\chi \rightarrow \neg\delta$

viii. δ

ix. $/ \therefore \neg\chi$

Which of these are *the* logical form of (1)-(3)? There are many different ways to capture its logical form. Which one of them has the right to be qualified as *the* logical form of (1)-(3)? This question is pressing if logical forms are taken to be schemata, and hence to be linguistic entities. If a logical form is just a string of symbols, then it varies by using a distinct set of variables. There will be no non-arbitrary way to choose one as opposed to any other as *the* logical form of a given argument. In other words, there will be nothing to choose between these linguistically distinct entities and, hence, none of them could be identified with *the* logical form of the original argument.

This may encourage us to identify logical forms as language-independent or language-invariant entities. On this view, logical forms are identified not with schemata, but with what schemata express or represent. They are worldly, rather than linguistic, entities. This view does not succumb to the above problem. Since, on this view, logical forms are worldly entities, none of the above candidates–i.e. (i)-(iii), (iv)-(vi), and (vii)-(ix)–is the logical form of (1)-(3). Rather, each of them *expresses* or *represents* its logical form.

ONE LOGICAL FORM OR MANY?

It seems then that we will be in a better position if we assume that logical forms are worldly entities. But this does not leave us completely home and dry, either. So far, we have assumed that logical forms are *unique* entities. That is, we assumed that arguments such as (1)-(3) and (7)-(9) have o*ne and the same* logical form. But is that the case?

In general, objects can take *many* forms. For example, a particular sonnet can be both Petrarchan and Miltonic, and a vase can be both a cuboid and a cube.[2] Also, it seems that a single sentence can take many (at least, more than one) forms. Consider $\neg(P \rightarrow \neg Q)$. What is its logical form? It seems that each of the following options works perfectly well as an answer to our question: it is a negation; it is a negation of a conditional; and it is a negation of a conditional whose consequent is a negation.[3]

Now, suppose that each of these logical forms is a logical form of a given argument. In virtue of what is each of them a logical form of one and the same argument? That is, what explains the fact that different logical forms are forms of one and the same argument? What unifies them in this respect? One answer is to say that all of these forms have a common logical form. But then you can ask the same question about this common logical form, since this very form has further different forms. In virtue of what are these logical forms forms of one and the same form? And this process can go endlessly. You have a logical form which itself has other logical forms, and so on. But this is not compatible with the thesis that logical forms are unique entities.[4]

2. See Oliver (2010, 172), where he disagrees with Strawson (195, 54).

3. This way of putting the point is due to Smith (2012, 81).

4. This is reminiscent of the Aristotelian Third Man argument against Plato's theory of Forms.

Question for Reflection

It seems that we cannot always talk of *the* logical form that an argument or various arguments share. If this view is correct, then what are its philosophical implications? Can we still understand the notion of validity in terms of the notion of logical form?

SUMMARY

This chapter started with a question about the subject matter of formal logic: what is it that formal logic studies? We discussed the thesis that formal logic studies logical consequence through the *form* of arguments. We then explicated the notion of validity in terms of truth-tables, which specify the conditions under which a proposition is true or false–for example, a conditional proposition is false only when its antecedent is true and its consequence false; otherwise, it is true. Thus, as we discussed above, truth-tables can be employed to determine whether arguments formulated in the language of propositional logic are valid.

We then dug further into what it means for arguments to have a logical form, and how their logical form impacts their (in)validity. The chief idea is that every argument which shares its logical form with a valid argument is also valid, and consequently, every argument which shares its logical form with an invalid argument is also invalid. We saw how this understanding of the notion of validity enables us to identify formal fallacies, such as the fallacy of affirming the consequent. We ended this chapter by asking three philosophical questions about the nature, existence, and uniqueness of logical forms.